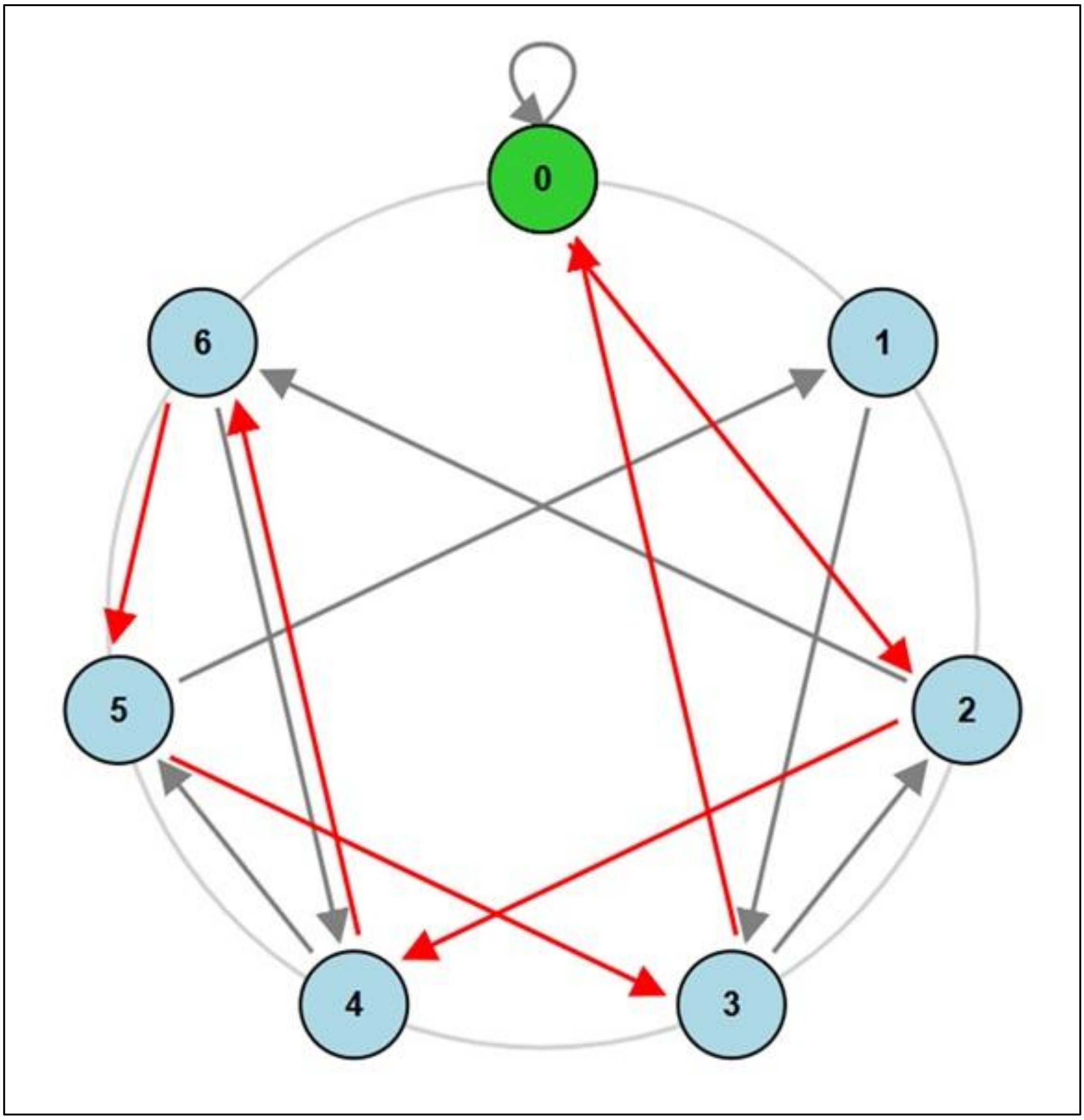
Universidad Rey Juan Carlos

DIVISIBILITY GRAPHS

Practice worksheet



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Suitability for different educational levels

The proposed exercises present different levels of difficulty and can therefore be adapted to different educational stages depending on students' knowledge and competencies.

In the final cycle of Primary Education, the simpler exercises from Parts 1 and 2 can be used, particularly those related to the construction and reading of divisibility graphs, performing simple calculations, and following sequences of states. For example, Exercises 1 and 7 allow students to work on divisibility and logical reasoning in a visual and progressive way.

In Secondary Education, a broader selection of activities can be addressed, including exercises that require greater analytical skills, justification, and pattern recognition. At this level, all the exercises in Parts 1 and 2 are particularly suitable.

Finally, for pre-university education and the first years of university, Part 3 presents exercises involving greater mathematical depth, focusing on abstract reasoning, the formulation and verification of conjectures, the generalization of results, and the study of patterns. These exercises also make it possible to connect the behavior of divisibility graphs with more advanced algebraic and arithmetic concepts.

In this way, the proposal can be understood as a resource with progressively increasing difficulty, allowing divisibility graphs to be introduced from an intuitive and visual perspective and gradually progressing towards mathematical analysis, generalization, and proof.

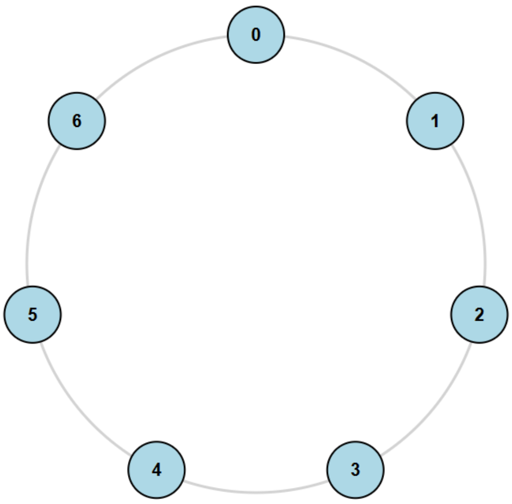
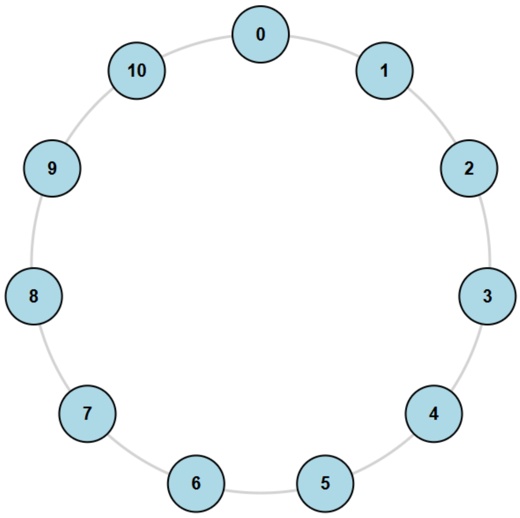
DIVISIBILITY GRAPHS

Practice worksheet

In this worksheet, you will find a series of exercises designed to help you work with and understand divisibility graphs. Solve them by reasoning through your answers. Use the available interactive resources to verify and review your answers once you have finished.

PART 1: CONSTRUCTION AND PROPERTIES OF DIVISIBILITY GRAPHS

Exercise 1. Construct the divisibility graphs corresponding to moduli 7, 11, and 13, indicating the calculations on which you based their construction.

<table>
<tr><td>Modulus 7

</td><td></td></tr>
<tr><td>Modulus 11

</td><td></td></tr>
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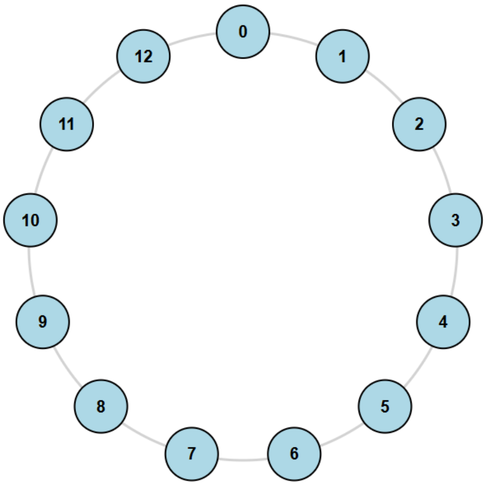
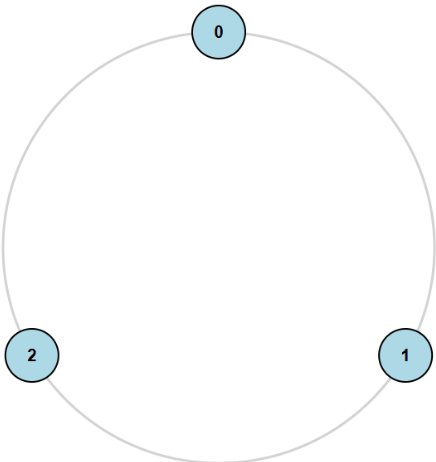
Modulus 13

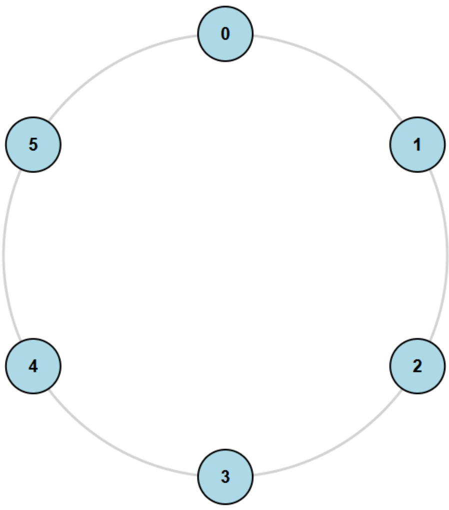


Exercise 2. Construct the divisibility graphs corresponding to moduli 3, 6, and 12, indicating the calculations on which you based their construction. What do you observe?

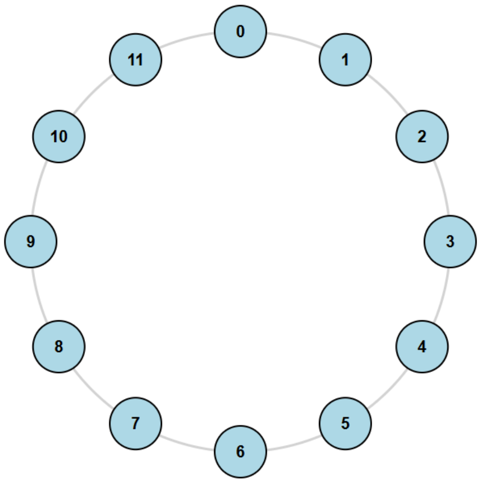
Modulus 3



Modulus 6

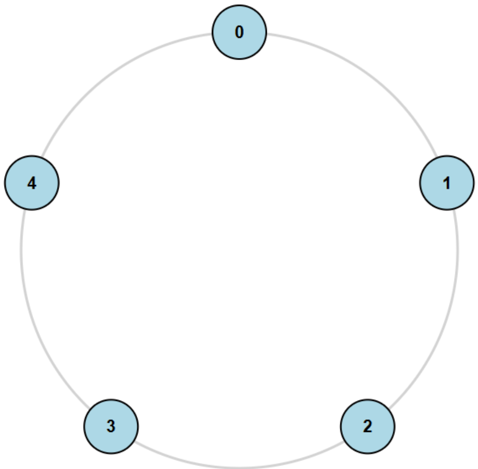


Modulus 12

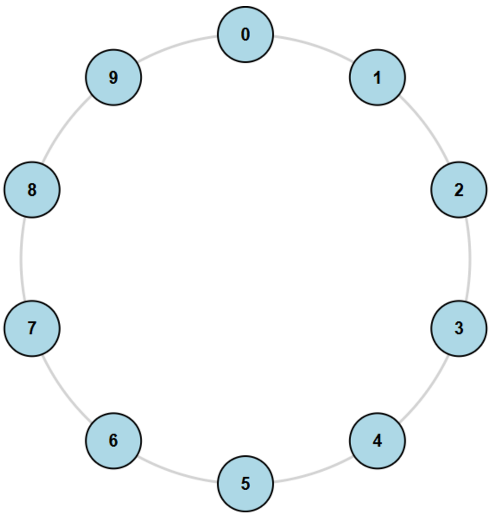


Exercise 3. Construct the divisibility graphs corresponding to moduli 5 and 10, indicating the calculations on which you based their construction. What do you observe?

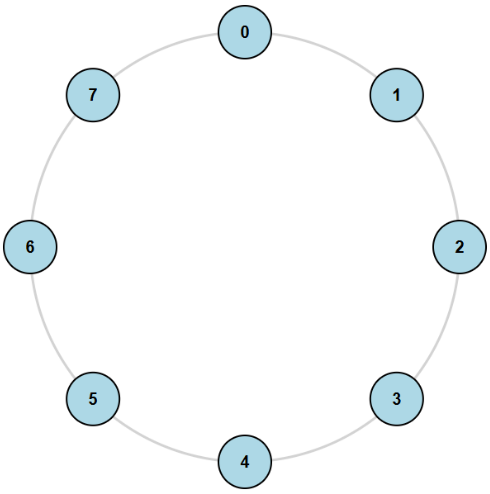
Modulus 5



Modulus 10



Exercise 4. Construct the divisibility graph modulo 8, indicating the calculations on which you based its construction. What do you observe between this graph and the divisibility graphs corresponding to moduli 4 and 12?

Modulus 8 	

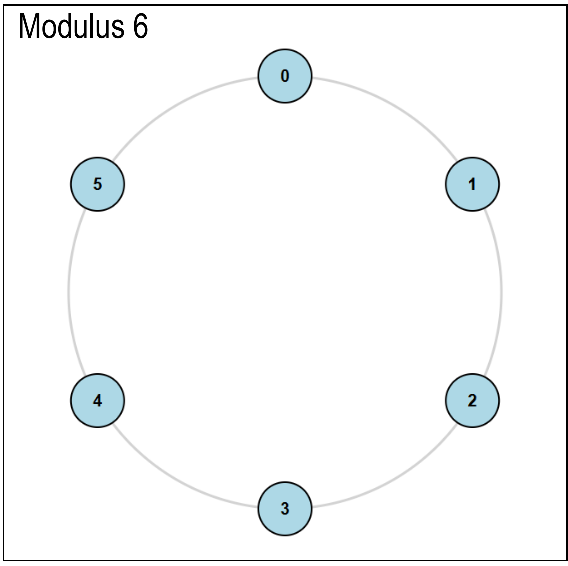
Exercise 5. Construct the divisibility graph modulo 15. What properties does it share with the divisibility graphs corresponding to moduli 3 and 5? Justify your answer.

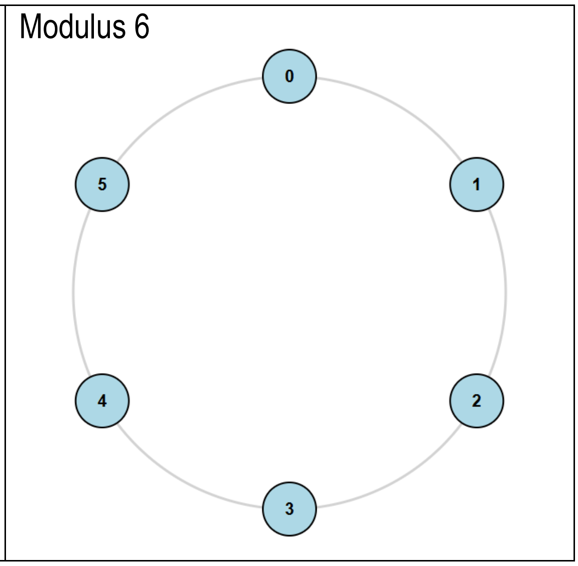
Exercise 6. What geometric property do all divisibility graphs modulo $m > 2$ have in common? Justify your answer.

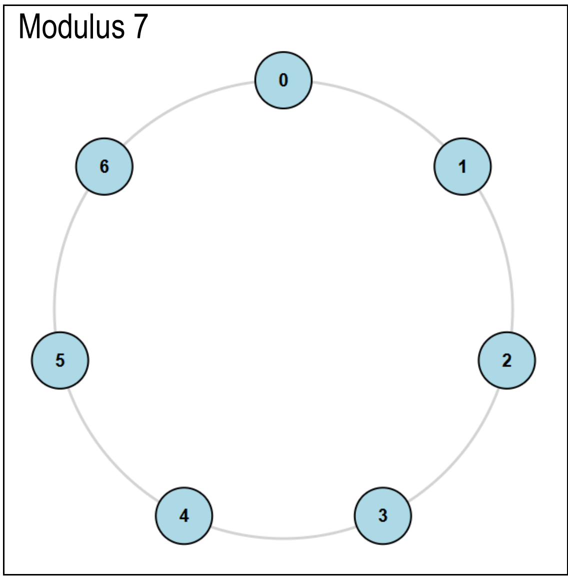
PART 2: ANALYSIS AND CONSTRUCTION OF REMAINDER WALKS

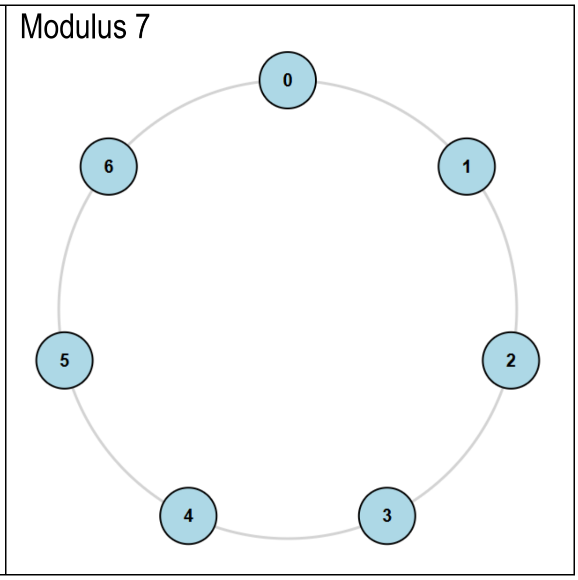
Exercise 7. Complete the following table:

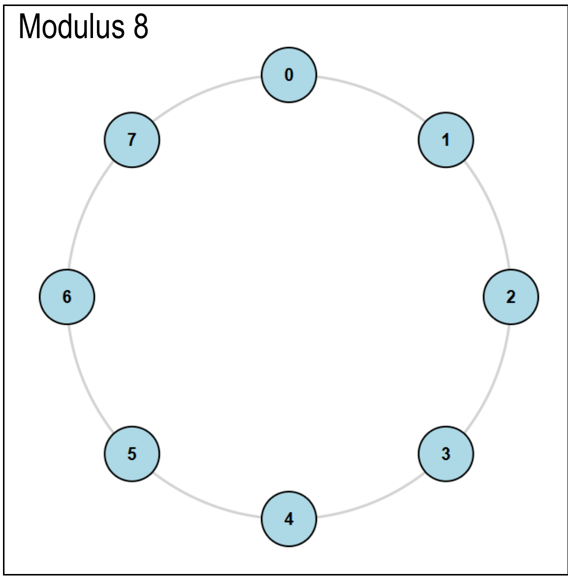
NUMBER	MODULUS	WALK	DIVISIBLE?
1234	11		
435071	7		
216561	8		
239274	9		
83472	6		
9208327	7		
136104	6		
100713	8		
40241	11		
232614	9		
422234	12		
283894	13		

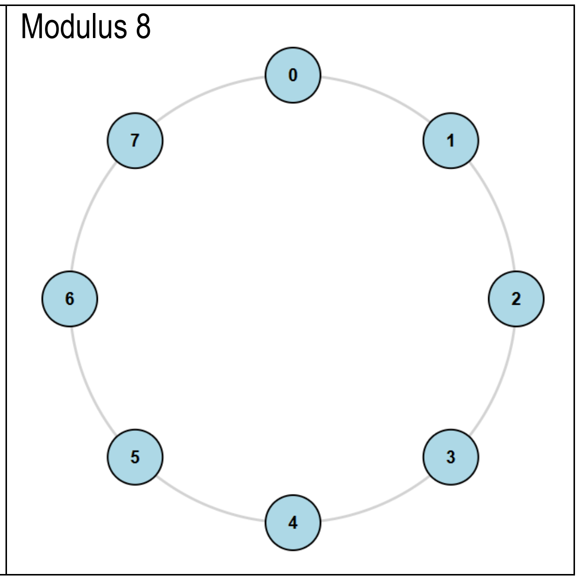

Modulus 6
0
1
2
3
4
5


Modulus 6
0
1
2
3
4
5

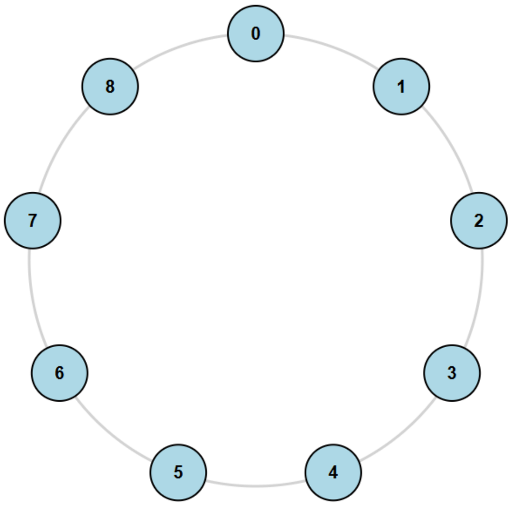

Modulus 7
0
1
2
3
4
5
6


Modulus 7
0
1
2
3
4
5
6

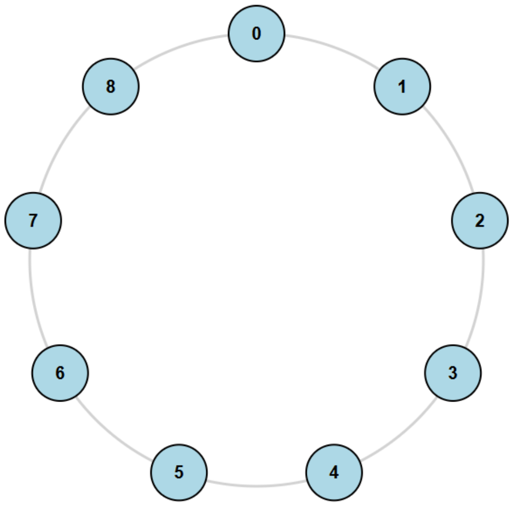

Modulus 8
0
1
2
3
4
5
6
7


Modulus 8
0
1
2
3
4
5
6
7

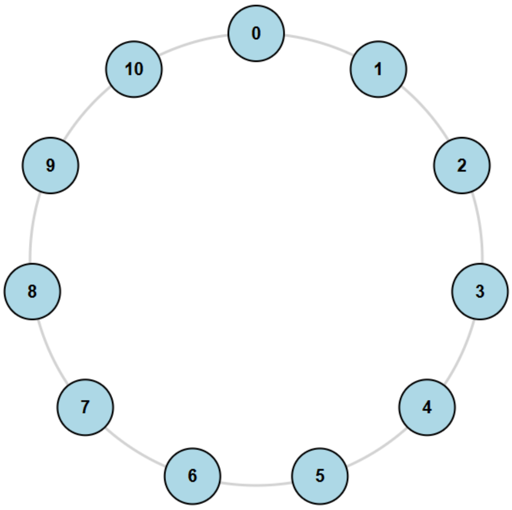
Modulus 9



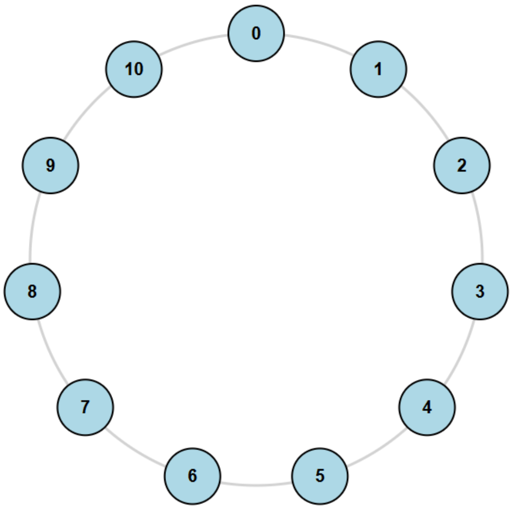
Modulus 9



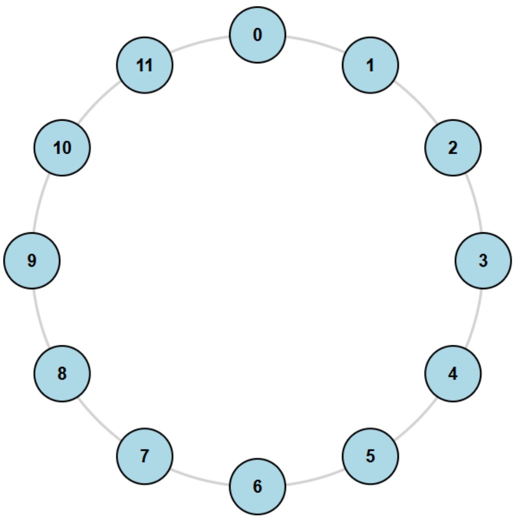
Modulus 11



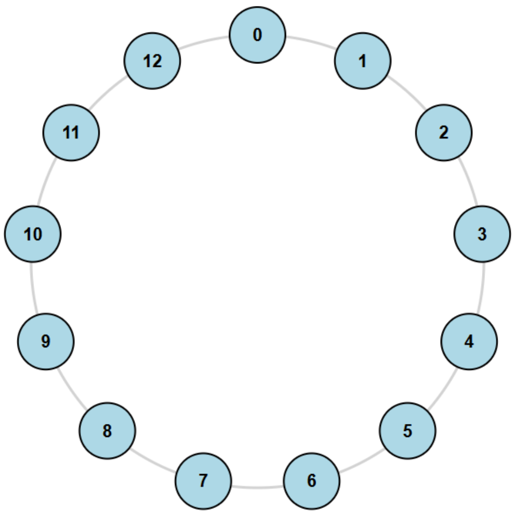
Modulus 11



Modulus 12



Modulus 13



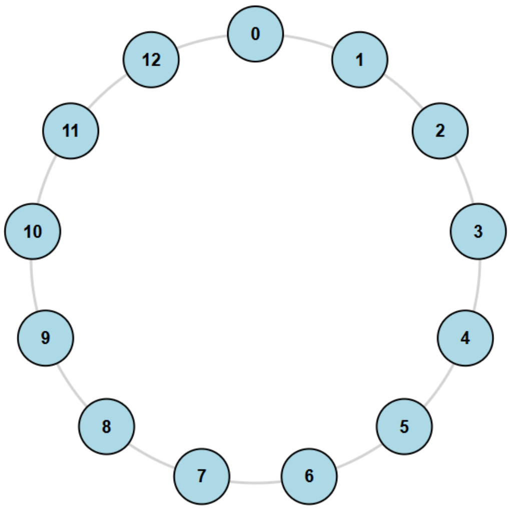
Exercise 8. Using the divisibility graph modulo 13, find a six-digit number with no zeros that is divisible by 13. Justify your answer.

Exercise 9. Using the divisibility graph modulo 7, find a seven-digit number with no zeros that is divisible by 7. Justify your answer.

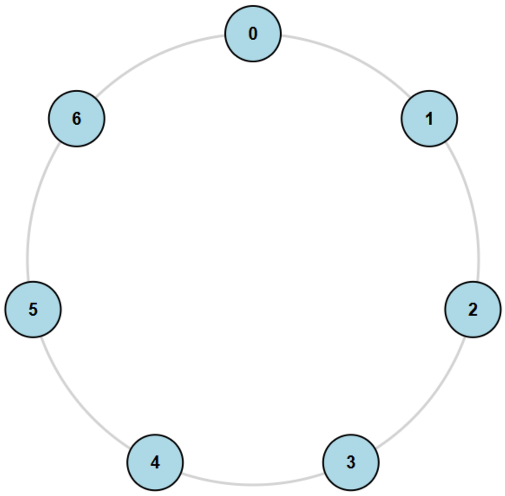
Exercise 10. Using the divisibility graph modulo 7, find a number that produces the following sequence of states: $0 \longrightarrow 5 \longrightarrow 4 \longrightarrow 0 \longrightarrow 1 \longrightarrow 4$.

Exercise 11. Using the divisibility graph modulo 11, find a number that produces the following sequence of states: $0 \longrightarrow 2 \longrightarrow 9 \longrightarrow 2 \longrightarrow 0 \longrightarrow 0$.

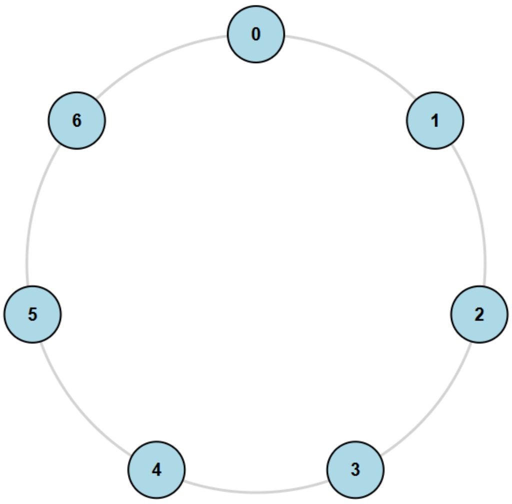
Exercise 8



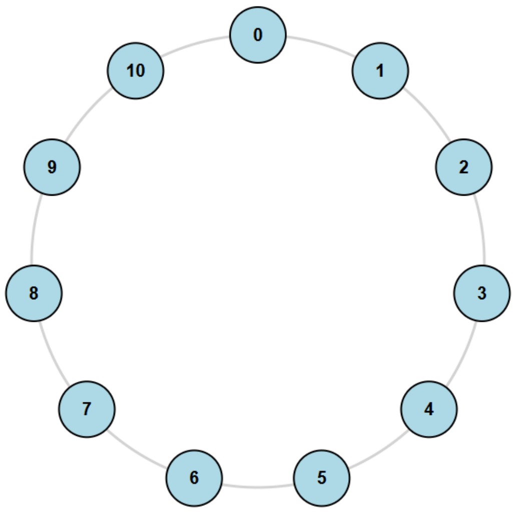
Exercise 9



Exercise 10



Exercise 11



Exercise 12. In a divisibility graph modulo 13, is it possible for the sequence of states generated by a number to move from state 2 to state 4? Justify your answer.

Exercise 13. Consider the divisibility graph modulo 6. Which numbers, with a non-zero first digit, are processed by the following sequence of states: 0 ⟶ 0 ⟶ 0 ⟶ 0?

Exercise 14. Find all the numbers generated by the walk in Exercise 12.

Exercise 15. When the number 2163 is processed through a divisibility graph modulo m, the following walk is obtained: 0 ⟶ 2 ⟶ 9 ⟶ 0 ⟶ 3. Find the value of m.

Exercise 16. When the number 2727 is processed through a divisibility graph modulo m, the following walk is obtained: 0 ⟶ 2 ⟶ 0 ⟶ 2 ⟶ 0. Given that $2 \leq m \leq 9$, which values of m make this walk possible?

Exercise 17. When a natural number is processed through the divisibility graph modulo 5, the following walk is obtained: 0 ⟶ 1 ⟶ 1 ⟶ 1 ⟶ 1. How many natural numbers produce this walk? Among these numbers, is there any that contains repeated digits? Is there any that contains the digit 5? Justify your answer.

Exercise 18. Let $N_1 = 702^3 \times 395^2 + 1482$ and $N_2 = 4178 + 3942^k$. Using the divisibility graph modulo 7, prove that: (a) N_1 is divisible by 7; (b) N_2 is divisible by 7 for every integer $k \geq 0$.

PART 3: ADVANCED ANALYSIS – EXPLORATION AND REASONING

Exercise 19. Consider the divisibility graph modulo 8. The sequence of states that processes the number $N = a_1 3 a_3 7 a_5 a_6$ is:

$$0 \longrightarrow 2 \longrightarrow r_2 \longrightarrow 6 \longrightarrow r_4 \longrightarrow 0 \longrightarrow 0,$$

where $0 \leq r_2, r_4 \leq 7$. Find the values of the digits a_1, a_3, a_5 y a_6 such that the number N is divisible by 8.

Exercise 20. Let $N = 847^k - 692$. Using the divisibility graph modulo 13, find the values of k for which N is divisible by 13.

Exercise 21. Determine whether the following statement is true or false: «Let there be a divisibility graph modulo $m \geq 2$ and a natural number whose first digit is $a_1 > 0$. The sequence of states will always begin with $0 \longrightarrow a_1$, regardless of the value of m». Justify your answer.

Exercise 22. Determine whether the following statement is true or false: «Let D_m be a divisibility graph modulo m, with $2 \leq m \leq 9$, and let R be a previously defined walk. The number processed by R through the graph G is unique».

Exercise 23. Determine whether the following statement is true or false: «Let D_m be a divisibility graph modulo $m \geq 10$, and let R be a previously defined walk. The number processed by R through the graph G is unique».

Exercise 24. Consider the divisibility graph modulo 5. Suppose that, when a natural number is processed, the following walk is obtained:

$$0 \to k \to k \to (\cdots) \to k,$$

If the walk contains n transitions, determine the possible digits that can produce each transition. How many n-digit natural numbers produce this walk?

Exercise 25. Consider the divisibility graph modulo 3. Walks of the form:

$$0 \to k \to 0 \to k \to 0 \to (\cdots) \to 0 \to k \to 0,$$

donde $k \in \{1, 2\}$, produce numbers whose digits contain neither 3, 6, nor 9. Why do you think this happens? Justify your answer.

Exercise 26. Determine whether the following statement is true or false: «Let D_m be a divisibility graph modulo m. Let N be a natural number with n-digit, defined by:

$$N = m10^{n-1},$$

that is, the n-digit number $(m00 \cdots 0)$. The sequence of states, of length n, generated by N will always be:

$$0 \to 0 \to 0 \to (\cdots) \to 0,$$

for every $m \in \{2, 3, \dots, 9\}$». Justify your answer.

Exercise 27. In exercise 21, the statement was studied for $m \in \{2, 3, \dots, 9\}$. Determine whether the same statement remains true when m successively belongs to the intervals:

$$\{10, 11, \dots, 19\}, \ \ \{20, 21, \dots, 29\}, \ \ \dots, \ \ \{90, 91, \dots, 99\}.$$

That is, for each interval, analyze the sequence of states generated by the numbers:

$$N(m, n) = m10^{n-2}$$

and justify your conclusions.

Exercise 28. Let D_m be a divisibility graph modulo m, with $10 \le m \le 19$. Let N be the natural number with $2n$-digit obtained by concatenating n copies of m. That is,

$$N(m, n) = \sum_{i=0}^{n-1} m \cdot 10^{2i} = m\frac{10^{2n}-1}{99},$$

For example, $N(12, 5) = 1212121212$. Study the form of the sequence of states generated by N, for every $m \in \{10, 11, \dots, 19\}$. Is N divisible by m? Justify your answer.

Exercise 29. Considering the situation described in Exercise 23, extend the study to the following intervals:

$$30 \le m \le 39, \quad 40 \le m \le 49, \quad \dots, \quad 90 \le m \le 99.$$

For each interval, study the form of the sequence of states generated by N. Which patterns remain unchanged, and what changes occur as the interval of values of m increases? Justify your answer.

Exercise 30. Let D_m be a divisibility graph modulo m, with $2 \leq m \leq 10$. Let N be a natural number with n-digit, defined by:

$$N = \sum_{i=1}^{n} 10^{n-i} a_i, \quad 1 \leq a_1 \leq 9, \quad 0 \leq a_i \leq 9,$$

that is, the n-digit number $(a_1 a_2 \cdots a_n)$. When N is processed through the graph D_m, the following walk is obtained:

$$0 \rightarrow r_1 \rightarrow r_2 \rightarrow (\cdots) \rightarrow r_n, \quad 0 \leq r_i \leq 9.$$

Determine the values of m for which $a_i = r_i$, for every $i \in \{1, 2, 3, \dots, n\}$. In other words, find the values of m for which the walk obtained from processing N coincides exactly with the sequence of digits of N. Justify your answer.